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Inventor(s)	Wagner; Hannah Elizabeth et al.

Door handle locator system for a vehicle

Abstract

A door handle locator system for a vehicle includes one or more modalities that provide sensory feedback to a user indicating a location of a door handle that is part of a door of the vehicle. The door handle locator system also includes one or more controllers in electronic communication with the one or more modalities. The one or more controllers execute instructions to receive a wireless signal indicating a location of the user with respect to the vehicle. The one or more controllers determine the user is approaching the vehicle based on the location of the user with respect to the vehicle. In response to determining the user is approaching the vehicle, the one or more controllers instruct the one or more modalities to provide the sensory feedback to the user indicating the location of the door handle.

Inventors: Wagner; Hannah Elizabeth (Berkley, MI), Bond; Jacob Alan (Rochester Hills, MI), Szczerba; Joseph F. (Grand Blanc, MI), Rajavenkatanarayanan; Akilesh (Macomb, MI)

Applicant: GM Global Technology Operations LLC (Detroit, MI)

Family ID: 94213010

Assignee: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

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Primary Examiner: Barakat; Mohamed

Attorney, Agent or Firm: Vivacqua Crane, PLLC

Background/Summary

INTRODUCTION

- (1) The present disclosure relates to a door handle locator system for a vehicle that includes one or more modalities that provide sensory feedback to a user indicating the location of a door handle of the vehicle.
- (2) An individual may actuate a door handle to gain entry or to exit a vehicle. Vehicles include interior door handles that allow passengers to exit the vehicle as well as exterior door handles that allow individuals to gain access to the interior cabin of the vehicle. Some individuals may find it challenging to locate the door handles during low-light conditions such as, for example, at dusk or night. It is to be appreciated that individuals who are visually impaired may have an especially difficult time locating the door handle of a vehicle during low-light conditions because of their limited vision capabilities. Moreover, it is also to be appreciated that the door handles of a vehicle are designed to blend in with the vehicle for aesthetic purposes, which exacerbates the issues individuals may have with locating the door handles during low-light conditions.
- (3) Thus, while door handles for vehicles achieve their intended purpose, there is a need in the art for an approach to assist an individual with locating a door handle.

SUMMARY

- (4) According to several aspects, a door handle locator system for a vehicle, where the vehicle

includes a door having a door handle. The door handle locator system includes one or more modalities that provide sensory feedback to a user indicating a location of the door handle, and one or more controllers in electronic communication with the one or more modalities. The one or more controllers execute instructions to receive a wireless signal indicating a location of the user with respect to the vehicle. The one or more controllers execute instructions to determine the user is approaching the vehicle based on the location of the user with respect to the vehicle. In response to determining the user is approaching the vehicle, the one or more controllers instruct the one or more modalities to provide the sensory feedback to the user indicating the location of the door handle.

(5) In another aspect, the one or more modalities include one or more haptic mechanisms that generate haptic feedback.

(6) In yet another aspect, the one or more haptic mechanisms create a vibration pattern that includes a plurality of individual vibrations that are dissipated throughout a body of the door, where the user locates the door handle based on an intensity of the individual vibrations.

(7) In an aspect, the one or more haptic mechanisms are located within a body of the door.

(8) In another aspect, the one or more haptic mechanisms are positioned within the door handle.

(9) In another aspect, the one or more controllers modulate at least one of the following: an intensity, a length, and a pitch of the haptic feedback to convey information regarding one or more of the following: the door handle, the door opened by the door handle, and an interior seat of the vehicle that corresponds to the door opened by the door handle.

(10) In yet another aspect, the one or more modalities include one or more light-emitting elements that emit visible light.

(11) In an aspect, the one or more controllers modulate at least one of the following: an intensity and color of the visible light to convey information regarding one or more of the following: the door handle, the door opened by the door handle, and an interior seat of the vehicle that corresponds to the door opened by the door handle.

(12) In another aspect, the one or more controllers instruct the one or more light-emitting elements to create animation to indicate one or more of the following: a specific location of the door handle relative to the door, a door status of the door opened by the door handle, and an occupancy status of an interior seat that corresponds to the door.

(13) In yet another aspect, the one or more modalities include one or more speakers that generate sound that is audible to humans.

(14) In an aspect, the one or more controllers modulate at least one of the following: a frequency and an intensity of the sound emitted by the one or more speakers to guide the user towards the door handle.

(15) In another aspect, the sound emitted by the one or more speakers includes voice instructions.

(16) In yet another aspect, the one or more modalities include one or more actuation elements that actuate the door handle between a retracted position and a deployed position.

(17) In an aspect, the one or more controllers instruct the one or more actuation elements to rotate the door handle from the retracted position to the deployed position in response to determining the user is approaching the vehicle.

(18) In another aspect, a distal end of the door handle projects outwards relative to an outer surface of the door when the door handle is in the deployed position.

(19) In yet another aspect, the one or more modalities include one or more electromagnets that create a magnetic field when provided with an electric current.

(20) In an aspect, the user wears a complimentary magnet that includes a polarity that is opposite of a polarity of the one or more electromagnets.

(21) In another aspect, a magnetic force is created as the one or more electromagnets attract the complimentary magnet, and wherein the magnetic force is of a magnitude sufficient to urge a hand of the user towards the door handle of the vehicle.

(22) In yet another aspect, a method for indicating a location of a door handle for a vehicle by a door handle locator system. The method includes receiving, by one or more controllers, a wireless signal indicating a location of a user with respect to the vehicle. The method includes determining, by the one or more controllers, a user is approaching the vehicle based on the location of the user with respect to the vehicle. In response to determining the user is approaching the vehicle, the method includes instructing one or more modalities to provide sensory feedback to the user indicating the location of the door handle.

(23) In an aspect, a door handle locator system for a vehicle, where the vehicle includes a door having a door handle. The door handle locator system includes one or more modalities that provide sensory feedback to a user indicating a location of the door handle, where the one or more modalities include one or more of the following: one or more haptic mechanisms, one or more light-emitting elements, one or more speakers, one or more actuation elements, and one or more electromagnets. The door handle locator system includes one or more controllers in electronic communication with the one or more modalities. The one or more controllers execute instructions to receive a wireless signal indicating a location of the user with respect to the vehicle, determine the user is approaching the vehicle based on the location of the user with respect to the vehicle, and in response to determining the user is approaching the vehicle, instruct the one or more modalities to provide the sensory feedback to the user indicating the location of the door handle.

(24) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

(2) FIG. 1 is a schematic diagram of a vehicle including the disclosed door handle locator system including one or more controllers in electronic communication with one or more modalities that provide sensory feedback to a user indicating the location of a door handle, according to an exemplary embodiment;

(3) FIG. 2A illustrates one of the doors of the vehicle including one or more haptic mechanisms as the modality, according to an exemplary embodiment;

(4) FIG. 2B illustrates a vibrational pattern that is created along the door of the vehicle by the haptic mechanism shown in FIG. 2A, according to an exemplary embodiment;

(5) FIG. 3A illustrates one of the doors of the vehicle including one or more light-emitting elements placed within the door handle as the modality, according to an exemplary embodiment;

(6) FIG. 3B illustrates one of the doors of the vehicle including one or more light-emitting elements placed around the door handle as the modality, according to an exemplary embodiment;

(7) FIG. 4A is a top view of one of the doors of the vehicle including one or more actuation elements as the modality, where the one or more actuation elements are in a retracted position, according to an exemplary embodiment;

(8) FIG. 4B is a top view of one of the doors of the vehicle including one or more actuation elements as the modality, where the one or more actuation elements are in a retracted position, according to an exemplary embodiment;

(9) FIG. 5 illustrates one of the doors of the vehicle including one or more electromagnets as the modality, according to an exemplary embodiment; and

(10) FIG. 6 is a process flow diagram illustrating a method for providing sensory feedback by the door handle locator system indicating the location of the door handle, according to an exemplary

embodiment.

DETAILED DESCRIPTION

(11) The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

(12) Referring to FIG. 1, a schematic diagram of an exemplary door handle locator system **10** for a vehicle **12** is illustrated. It is to be appreciated that the vehicle **12** may be any type of vehicle such as, but not limited to, a sedan, truck, sport utility vehicle, van, or motor home. As explained below, the door handle locator system **10** includes one or more modalities **14** that provide sensory feedback to a user **16** indicating the location of a door handle **18** that is part of the vehicle **12** when activated. The door handle locator system **10** includes one or more controllers **20** that are in electronic communication with the one or more modalities **14** and one or more door handles **18**. In the non-limiting embodiment as shown in FIG. 1, the one or more modalities **14** include one or more of the following: one or more haptic mechanisms **22**, one or more light-emitting elements **24**, one or more speakers **26**, one or more actuation elements **28**, and one or more electromagnets **30**. It is to be appreciated that although only a single door handle **18** is illustrated in FIG. 1, this is merely for purposes of simplicity and clarity, and each door **42** that is part of the vehicle **12** includes an interior door handle as well as an exterior door handle.

(13) The one or more controllers **20** of the door handle locator system **10** is in wireless communication with a personal electronic device **32**, where the personal electronic device **32** transmits a wireless signal to the one or more controllers **20** indicating a location of the user **16** with respect to the vehicle **12**. The one or more controllers **20** determine when the user **16** is approaching the vehicle **12** based on the location of the user **16** with respect to the vehicle **12**. For example, when the location of the user **16** is within a predefined distance from the vehicle **12**, the one or more controllers **20** determine the user **16** is approaching the vehicle **12**. In response to determining the user **16** is approaching the vehicle **12**, the one or more controllers **20** instruct the one or more modalities **14** to provide the sensory feedback to the user **16** indicating the location of the door handle **18**. The personal electronic device **32** is any type of electronic device having wireless communication capabilities that is portable so an individual may carry the personal electronic device **32** on his or her person. Some examples of the personal electronic device **32** include, but are not limited to, a key fob, a smartphone, and a smartwatch.

(14) In the event the door handle locator system **10** includes more than one modality **14**, in one embodiment the user **16** may choose which modalities **14** are activated to indicate the location of the door handle **18**. The user **16** may activate the modalities **14** by entering user input into an interface **40** that is part of the vehicle **12** or, alternatively, by the personal electronic device **32**. The interface **40** is any type of user interface that receives and processes user input such as, for example, an infotainment system. For example, if the personal electronic device **32** is a smartphone, then the user **16** may enter user input into the smartphone that activates the one or more light-emitting elements **24** as the modality **14** that is activated to indicate the location of the door handle **18**.

(15) In one embodiment, the user **16** enters input into either the personal electronic device **32** or the user interface **40** to customize the generate sensory feedback generated by the one or more modalities **14**. Specifically, the sensory feedback may be customized to only allow for specific modalities **14** corresponding to one or more of the door handles **18** that are part of the vehicle **12** to activate. For example, in one embodiment, the vehicle **12** is a ride-share vehicle. In this example, the haptic feedback is customized to allow for only the modalities **14** corresponding to the door handles **18** that provide access to the back seats of the vehicle **12** to activate, since ride-share passengers do not typically ride in the front seats of the vehicle **12**.

(16) FIG. 2A illustrates the one or more haptic mechanisms **22** shown in FIG. 1 for an exterior door handle **18** that is disposed along an exterior door **42** of the vehicle **12**. It is to be appreciated that while the figures illustrate the door handle **18** located on an exterior door **42** of the vehicle **12**, in

another embodiment the door handle **18** may be an interior door handle located within the interior cabin of the vehicle **12** as well. In the example as shown in FIG. 2A, the one or more haptic mechanisms **22** are illustrated in phantom line and are located within the body **44** of the door **42** and surround the door handle **18**. However, it is to be appreciated that in another embodiment the one or more haptic mechanisms **22** may be positioned within the door handle **18**. The one or more haptic mechanisms **22** include any device that generates haptic feedback such as, for example, eccentric rotating mass actuators.

(17) Referring to both FIGS. 2A and 2B, in response to determining the user **16** is approaching the vehicle **12**, the one or more controllers **20** (FIG. 1) instruct the one or more haptic mechanisms **22** to generate the sensor feedback, which is the haptic feedback. The haptic feedback creates a vibration pattern **46** that includes a plurality of individual vibrations **48** that dissipate throughout the body **44** of the door **42**. As seen in FIG. 2B, the individual vibrations **48** that are included as part of the vibration pattern **46** are stronger in intensity when proximate to the door handle **18**, where the intensity of the vibrations **48** gradually dissipates as the vibrations **48** travel throughout the body **44** of the door **42**. Therefore, the user **16** (FIG. 1) locates the door handle **18** based on the intensity of the individual vibrations **48** created by the one or more haptic mechanisms **22**.

(18) In one embodiment, an intensity, a length, a pitch, or a combination of intensity, length, or pitch of the haptic feedback generated by the one or more haptic mechanisms **22** is modulated to convey information regarding one or more of the following: the door handle **18**, the door **42** that is opened by the door handle **18**, and an interior seat of the vehicle **12** that corresponds to the door **42** that is opened by the door handle **18**. Specifically, in one embodiment, one or more of the intensity, the length, and the pitch of the haptic feedback is modulated to indicate a specific location of the door handle **18** relative to the door **42**, a door status of the door **42** that is opened by the door handle **18**, and an occupancy status of the interior seat that corresponds to the door **42**. The door status indicates if the door **42** is locked or unlocked, and the occupancy status indicates when a passenger is occupying a particular seat. For example, in one embodiment, the one or more haptic mechanisms **22** are modulated to generate repeated vibration patterns to indicate the specific location of the door handle **18** relative to the door **42**. As another example, the one or more haptic mechanisms **22** are modulated to generate three higher-pitched vibrations to indicate a seat is occupied.

(19) FIG. 3A illustrates the one or more light-emitting elements **24** shown in FIG. 1 for the door handle **18**. In the embodiment as shown in FIG. 3A, the light emitting-elements **24** are placed within the door handle **18**. Alternatively, in the embodiment as shown in FIG. 3B the light emitting elements **24** are located along an outer surface **60** of the door **42** and are positioned around the door handle **18**. The one or more light-emitting elements **24** include any device that generates visible light such as, for example, light-emitting diodes (LEDs). Referring to FIGS. 3A and 3B, in response to determining the user **16** is approaching the vehicle **12**, the one or more controllers **20** (FIG. 1) instruct the one or more light-emitting elements **24** to generate visible light. The visible light provides illumination and guides the hand of the user **16** towards the door handle **18**. In embodiments, the visible light is white light, however, it is to be appreciated that any other colored light may be used as well such as, for example, red light, blue light, or green light.

(20) In one embodiment, the one or more controllers **20** (FIG. 1) modulate the visible light emitted by the one or more light-emitting elements **24** to convey information regarding one or more of the following: the door handle **18**, the door **42** that is opened by the door handle **18**, and an interior seat of the vehicle **12** that corresponds to the door **42** that is opened by the door handle **18**. Specifically, in one embodiment, the one or more controllers **20** modulate the color, the intensity, or both the color and the intensity of the visible light to indicate a specific location of the door handle **18** relative to the door **42**, a door status of the door **42** that is opened by the door handle **18**, and an occupancy status of the interior seat that corresponds to the door **42**. For example, in one embodiment, to indicate the door status, the one or more controllers **20** may instruct the light-

emitting elements **24** generate blue light to indicate when the door **42** is unlocked, and white light to indicate when the door **42** is locked. In another example, the one or more controllers **20** instruct the light-emitting elements **24** to flash or illuminate to guide the user **16** towards a specific door handle **18** that corresponds to one of the doors **42** that are part of the vehicle **12**.

(21) In one embodiment, the one or more controllers **20** instruct the one or more light-emitting elements **24** to create animation to indicate a specific location of the door handle **18** relative to the door **42**, a door status of the door **42** that is opened by the door handle **18**, and an occupancy status of the interior seat that corresponds to the door **42**. It is to be appreciated that the light-emitting elements **24** animate by dynamically blinking, flashing, changing color, and changing intensity. For example, if a seat corresponding to one of the door handles **18** is occupied, then the one or more controllers **20** may instruct the light-emitting elements **24** to generate relatively short flashes of blue and white light to indicate the corresponding seat is occupied. In one embodiment, the color, the intensity, and the animation of the visible light emitted by the one or more light-emitting elements **24** is specifically customized to attract the attention of a guide dog or other service animal.

(22) Referring to FIG. **1**, in an embodiment the one or more modalities **14** include the one or more speakers **26**. The one or more speakers **26** may be positioned within the body **44** of one of the doors **42** (FIGS. **2A** and **2B**) in an area directly behind the door handle **18**. In response to determining the user **16** is approaching the vehicle **12**, the one or more controllers **20** (FIG. **1**) instruct the one or more speakers **26** to generate sound that is audible to humans. The sound guides the user **16** towards the door handle **18**. Specifically, in one embodiment, the one or more controllers **20** may guide the user **16** towards the door handle **18** by modulating the frequency, the intensity, or both the frequency and the intensity of the sound emitted by the one or more speakers **26**. In another embodiment, the sound emitted by the one or more controllers **20** includes voice instructions, where the voice instructions guide the user **16** towards the door handle **18**.

(23) In one embodiment, the one or more controllers **20** instruct the sound emitted by the one or more speakers **26** to convey information regarding one or more of the following: the door handle **18**, the door **42** that is opened by the door handle **18**, and an interior seat of the vehicle **12** that corresponds to the door **42** (FIG. **2A**) that is opened by the door handle **18**. In one embodiment, the one or more controllers **20** instruct the sound to indicate a door status of the door **42** that is opened by the door handle **18** or an occupancy status of the interior seat that corresponds to the door **42**. For example, voice instructions are used to indicate when a seat is occupied, such as voice instructions that state “seat is occupied, find another door”. As another example, an audio chime or voice instructions may be used to indicate when the door is locked, such as a voice instruction that states “the door is locked” or “the door is unlocked”. In another example, the one or more controllers **20** instruct the sound emitted by the speaker **26** to guide the user **16** towards a specific door handle **18** that corresponds to one of the doors **42** that are part of the vehicle **12**. For example, the voice instructions may state “incorrect door” if the user **16** attempts to open the passenger side door when the user **16** is hailing a ride-share vehicle that seats the user **16** in the back seat.

(24) FIGS. **4A** and **4B** illustrate a top view of the door **42** of the vehicle **12** including the one or more actuation elements **28**. Specifically, FIG. **4A** illustrates the door handle **18** in a retracted position and FIG. **4B** illustrates the door handle **18** in a deployed position, where the one or more actuation elements **28** actuate the door handle **18** between the retracted position and the deployed position. The one or more actuation elements **28** are any device that actuates the door handle **18** between the retracted position and the deployed position such as, but not limited to, an electric motor. The one or more actuation elements **28** are positioned within the body **44** of the door **42** and are operatively connected to a proximate end **62** of the door handle **18**.

(25) The door handle **18** is normally in the retracted position as seen in FIG. **4A**, where the proximate end **62** and a distal end **64** of the door handle **18** are positioned flush with the outer surface **60** of the door **42**. In response to determining the user **16** is approaching the vehicle **12**, the

one or more controllers **20** (FIG. **1**) instruct the one or more actuation elements **28** to rotate the door handle **18** from the retracted position as shown in FIG. **4A** to the deployed position shown in FIG. **4B**. When in the deployed position, the distal end **64** of the door handle **18** projects outwards relative to the outer surface **60** of the door **42** at an angle θ , thereby making the door handle **18** more prominent when compared to the retracted position. Accordingly, the user **16** (FIG. **1**) more easily views the door handle **18** when in the deployed position. In addition to making the door handle **18** more prominent, the door handle **18** is also easier for the user **16** to grasp as well, which may be especially advantageous for individuals who have issues with grasping or manipulating objects.

(26) FIG. **5** illustrates the one or more electromagnets **30** shown in FIG. **1** as the one or more modalities **14** for the door handle **18**. In the embodiment as shown in FIG. **5**, the one or more electromagnets **30** are placed within the body **44** of the door **42** directly adjacent to the door handle **18**, however, it is to be appreciated in alternative embodiments the electromagnets **30** may be placed within the door handle **18** as well. Referring to FIGS. **1** and **5**, in response to determining the user **16** is approaching the vehicle **12**, the one or more controllers **20** activate the one or more electromagnets **30** by providing an electric current to the one or more electromagnets **30** to create a magnetic field **M**.

(27) As seen in FIG. **5**, the user **16** wears a complimentary magnet **70**. The magnet **70** worn by the user **16** has a polarity that is opposite of the polarity of the one or more electromagnets **30** of the vehicle **12**. The complimentary magnet **70** is part of an accessory **72** worn on a hand **74** or forearm **76** of the user **16**. The accessory **72** may be a piece of jewelry such as, for example, a ring, bracelet, or watch band. Therefore, when the one or more electromagnets **30** are activated and the complimentary magnet **70** worn by the user **16** enters the magnetic field **M**, a magnetic force is created as the one or more electromagnets **30** attract the complimentary magnet **70** worn by the user **16**. The magnetic force is of a magnitude sufficient to urge the hand **74** of the user **16** in the direction towards the door handle **18** of the vehicle **12**. That is, in other words, the magnetic force is of a magnitude strong enough so the user **16** detects a force that is guiding his or her hand **74** towards the door handle **18**.

(28) FIG. **6** is a process flow diagram illustrating an exemplary method **600** for providing sensory feedback to the user **16** by the door handle locator system **10** indicating the location of the door handle **18** that is part of the vehicle **12**. Referring to FIGS. **1** and **6**, the method **600** may begin at block **602**. In block **602**, the one or more controllers **20** receive a wireless signal indicating the location of the user **16** with respect to the vehicle **12** from the personal electronic device **32**. The method **600** may then proceed to block **604**.

(29) In decision block **604**, the one or more controllers **20** determine if the user **16** is approaching the vehicle **12** based on the location of the user **16** with respect to the vehicle **12**, which is indicated by the wireless signal. For example, when the location of the user **16** is within a predefined distance from the vehicle **12**, the one or more controllers **20** determine the user **16** is approaching the vehicle **12**. In response to determining the user **16** is not approaching the vehicle **12**, the method **600** terminates. However, in response to determining the user **16** is approaching the vehicle **12**, the method **600** proceeds to block **606**.

(30) In block **606**, in response to determining the user **16** is approaching the vehicle **12**, the one or more controllers **20** instruct the one or more modalities **14** to provide the sensory feedback to the user **16** indicating the location of the door handle **18**. It is to be appreciated that the user **16** may select which modalities **14** are activated to indicate the location of the door handle **18**. The method **600** may then terminate.

(31) Referring generally to the figures, the disclosed door handle locator system provides various technical effects and benefits. Specifically, the disclosed door handle locator system provides an approach for providing sensory feedback indicating a location of the door handle of the vehicle. It is to be appreciated that the sensory feedback provided by the one or more modalities may be

especially helpful for individuals who are visually impaired and have an especially difficult time locating the door handle of a vehicle during low-light conditions. In addition to the location of the door handle, the door handle locator system may also provide other information to an individual as well, such as information regarding the door status and the occupancy status of the interior seat that corresponds to the door.

(32) The controllers may refer to, or be part of an electronic circuit, a combinational logic circuit, a field programmable gate array (FPGA), a processor (shared, dedicated, or group) that executes code, or a combination of some or all of the above, such as in a system-on-chip. Additionally, the controllers may be microprocessor-based such as a computer having at least one processor, memory (RAM and/or ROM), and associated input and output buses. The processor may operate under the control of an operating system that resides in memory. The operating system may manage computer resources so that computer program code embodied as one or more computer software applications, such as an application residing in memory, may have instructions executed by the processor. In an alternative embodiment, the processor may execute the application directly, in which case the operating system may be omitted.

(33) The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A door handle locator system for a vehicle, wherein the vehicle includes a door having a door handle, the door handle locator system comprising: one or more modalities that provide sensory feedback to a user indicating a location of the door handle, wherein the one or more modalities include one or more haptic mechanisms that generate haptic feedback, and wherein the one or more haptic mechanisms create a vibration pattern that includes a plurality of individual vibrations that are dissipated throughout a body of the door, and wherein the user locates the door handle based on an intensity of the individual vibrations; and one or more controllers in electronic communication with the one or more modalities, wherein the one or more controllers execute instructions to: receive a wireless signal indicating a location of the user with respect to the vehicle; determine the user is approaching the vehicle based on the location of the user with respect to the vehicle; and in response to determining the user is approaching the vehicle, instruct the one or more modalities to provide the sensory feedback to the user indicating the location of the door handle.

2. The door handle locator system of claim 1, wherein the one or more haptic mechanisms are located within a body of the door.

3. The door handle locator system of claim 1, wherein the one or more haptic mechanisms are positioned within the door handle.

4. The door handle locator system of claim 1, wherein the one or more controllers modulate at least one of the following: an intensity, a length, and a pitch of the haptic feedback to convey information regarding one or more of the following: the door handle, the door opened by the door handle, and an interior seat of the vehicle that corresponds to the door opened by the door handle.

5. The door handle locator system of claim 1, wherein the one or more modalities include one or more light-emitting elements that emit visible light.

6. The door handle locator system of claim 5, wherein the one or more controllers modulate at least one of the following: an intensity and color of the visible light to convey information regarding one or more of the following: the door handle, the door opened by the door handle, and an interior seat of the vehicle that corresponds to the door opened by the door handle.

7. The door handle locator system of claim 5, wherein the one or more controllers instruct the one or more light-emitting elements to create animation to indicate one or more of the following; a

specific location of the door handle relative to the door, a door status of the door opened by the door handle, and an occupancy status of an interior seat that corresponds to the door.

8. The door handle locator system of claim 1, wherein the one or more modalities include one or more speakers that generate sound that is audible to humans.

9. The door handle locator system of claim 8, wherein the one or more controllers modulate at least one of the following: a frequency and an intensity of the sound emitted by the one or more speakers to guide the user towards the door handle.

10. The door handle locator system of claim 8, wherein the sound emitted by the one or more speakers includes voice instructions.

11. The door handle locator system of claim 1, wherein the one or more modalities include one or more actuation elements that actuate the door handle between a retracted position and a deployed position.

12. The door handle locator system of claim 11, wherein the one or more controllers instruct the one or more actuation elements to rotate the door handle from the retracted position to the deployed position in response to determining the user is approaching the vehicle.

13. The door handle locator system of claim 11, wherein a distal end of the door handle projects outwards relative to an outer surface of the door when the door handle is in the deployed position.

14. The door handle locator system of claim 1, wherein the one or more modalities include one or more electromagnets that create a magnetic field when provided with an electric current.

15. The door handle locator system of claim 14, wherein the user wears a complimentary magnet that includes a polarity that is opposite of a polarity of the one or more electromagnets.

16. The door handle locator system of claim 15, wherein a magnetic force is created as the one or more electromagnets attract the complimentary magnet, and wherein the magnetic force is of a magnitude sufficient to urge a hand of the user towards the door handle of the vehicle.

17. A method for indicating a location of a door handle for a vehicle by a door handle locator system, the method comprising: receiving, by one or more controllers, a wireless signal indicating a location of a user with respect to the vehicle; determining, by the one or more controllers, a user is approaching the vehicle based on the location of the user with respect to the vehicle; and in response to determining the user is approaching the vehicle, instruct one or more modalities to provide sensory feedback to the user indicating the location of the door handle, wherein the one or more modalities include one or more haptic mechanisms that generate haptic feedback, and wherein the one or more haptic mechanisms create a vibration pattern that includes a plurality of individual vibrations that are dissipated throughout a body of the door, and wherein the user locates the door handle based on an intensity of the individual vibrations.

18. A door handle locator system for a vehicle, wherein the vehicle includes a door having a door handle, the door handle locator system comprising: one or more modalities that provide sensory feedback to a user indicating a location of the door handle, wherein the one or more modalities include one or more actuation elements, and one or more electromagnets that create a magnetic field when provided with an electric current, and wherein the user wears a complimentary magnet that includes a polarity that is opposite of a polarity of the one or more electromagnets, and wherein a magnetic force is created as the one or more electromagnets attract the complimentary magnet, and wherein the magnetic force is of a magnitude sufficient to urge a hand of the user towards the door handle of the vehicle; and one or more controllers in electronic communication with the one or more modalities, wherein the one or more controllers execute instructions to: receive a wireless signal indicating a location of the user with respect to the vehicle; determine the user is approaching the vehicle based on the location of the user with respect to the vehicle; and in response to determining the user is approaching the vehicle, instruct the one or more modalities to provide the sensory feedback to the user indicating the location of the door handle.

19. The door handle locator system of claim 18, wherein the one or more modalities further include

one or more of the following: one or more haptic mechanisms, one or more light-emitting elements, one or more speakers, and one or more actuation elements.
